LMU LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN
LMU MUNICH SCHOOL OF MANAGEMENT
INSTITUT FÜR RECHNUNGSWESEN UND WIRTSCHAFTSPRÜFUNG

Empirical Research in Accounting, Auditing and Capital Markets
Tutorial 3 · Measurement

Winter semester 2026/27 · M.Sc. / MBR

Dr. Jan Seitz

Room 316 and Zoom, 5 November 2026

TUTORIAL 3 · THURSDAY, 5 NOVEMBER 2026, 16:00–19:00

Measurement and Construct Validity

I. Concepts and proxies: Libby boxes for Lang & Lundholm
II. Describing data: tables and figures
III. Proxies and construct validity
IV. Design exercise: Lang & Lundholm today
V. Wrap-up

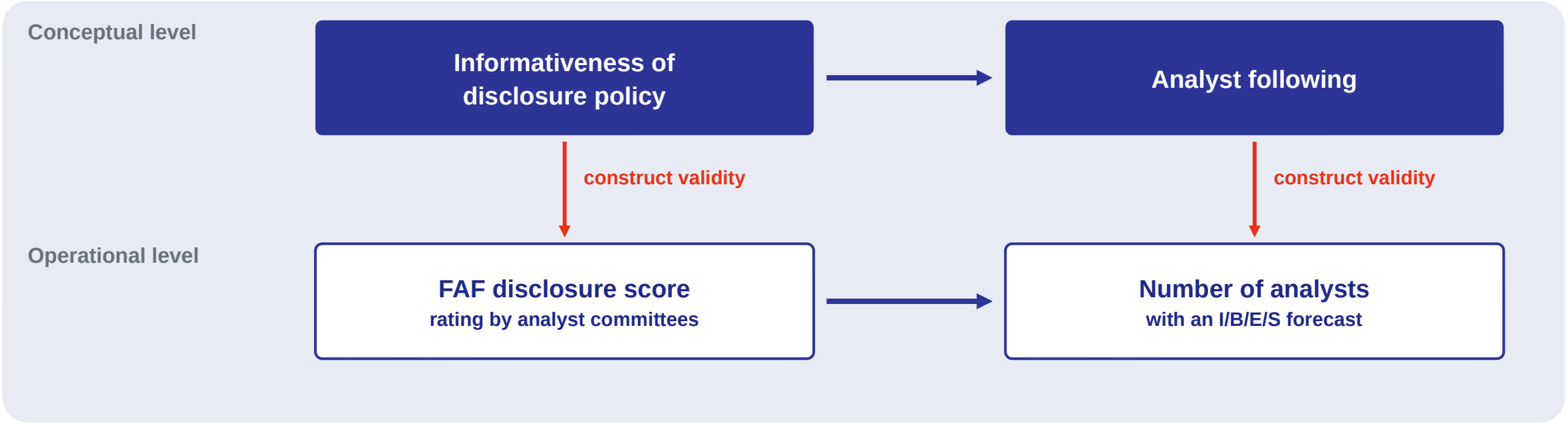
Today at a glance

Time	Block	What you do	R Markdown
16:00	I · Concepts and proxies	Libby boxes for Lang & Lundholm (1996) and Huang et al. (2026)	Part 1
16:25	II · Describing data	Descriptive table, ggplot2, winsorizing vs. truncating	Parts 2–3
17:30	Break	10 minutes	
17:40	III · Proxies and construct validity	Size proxies, scaling and ranks, SRN completeness	Part 4
18:25	IV · Design exercise	Lang & Lundholm today — in pairs	Part 5
18:50	V · Wrap-up	Checklist and outlook	Part 6

▶️ **IN RSTUDIO** **Pull the new material and open session_03/Session_03_student.Rmd**

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Every study translates concepts into measures



Other influences: firm size · return volatility · earnings–return correlation — L&L control for them and rank all variables within industry-years

ROOM AND ZOOM CHAT: WHICH ARROW IS THE WEAKEST — AND WHY?

Lang, M. H., & Lundholm, R. J. (1996). Corporate disclosure policy and analyst behavior. The Accounting Review, 71(4), 467–492, pp. 473–477.

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Same concept, different proxies: forecast accuracy

Lang & Lundholm (1996)

- Accuracy = − |EPS − median forecast| / price
- Firm level: consensus of all analysts
- Price scaling: comparable across firms — but tiny prices explode the ratio

Huang, Hugon, Zhang & Zheng (2026)

- Accuracy = − PMAFE: analyst's error / average error of all analysts for that firm
- Analyst level: compares analysts with each other
- X = AI_Prob from a machine-learning detector

A construct-validity threat the authors discuss themselves

“Many of the linguistic features used to detect AI-assisted reports are similar to features that proxy for writing quality” — so AI_Prob might measure good writers, not AI use.

▶️ **IN RSTUDIO** **Part 1 · Task A: Libby boxes for Huang et al. (2026) — which link worries you most?**

Huang, S., Hugon, A., Zhang, T., & Zheng, W. (2026). Generative AI and investment research: Evidence from analyst reports. Working paper, SSRN 6682498, pp. 1–5, 14.

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Describe before you analyse: Table 1 of every paper

What a descriptive table shows

- **N** — how many observations (and how many missing)
- **Mean vs. median** — skewness: a few giants pull the mean
- **SD, p25, p75** — typical spread
- **p1, p99** — extreme values and possible errors
- Units! (USD millions, employees in thousands)

Chunk 2.3

```
cs |>
  select(mv, at, sale, emp, roa) |>
  pivot_longer(everything()) |>
  group_by(name) |>
  summarise(
    N      = sum(!is.na(value)),
    mean   = mean(value, na.rm = TRUE),
    median = median(value, na.rm = TRUE),
    p1  = quantile(value, .01, na.rm = TRUE),
    p99 = quantile(value, .99, na.rm = TRUE))
```

▶️ **IN RSTUDIO** — **Part 2 · Chunks 2.1–2.4 + Task B: build variables, frequencies, descriptive table — read it like a referee**

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ggplot2: data + aesthetics + geoms

Chunk 3.1

```
ggplot(cs, aes(x = log(mv))) +
  geom_histogram(bins = 50) +
  labs(title = "Log market value",
       x = "log(MV)", y = "Firm-years")
```

You want to see …	geom
a distribution	geom_histogram()
groups compared	geom_boxplot()
a relation	geom_point() + geom_smooth()
a trend over time	geom_line()

Extreme values: three options

- **Log** — for positive, right-skewed variables (size)
- **Winsorize** at 1% / 99% — replace extremes by the percentile, keep the observation
- **Truncate** — delete extremes (changes the sample!)
- Whatever you choose: report it, and check that results do not hinge on it

▶️ **IN RSTUDIO** — **Part 3 · Chunk 3.1: histograms of market value — raw and log**

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Winsorizing or truncating?

	Winsorizing	Truncating
Extreme values are …	replaced by the 1% / 99% percentile	deleted
Sample size	unchanged	smaller
Risk	capped values are not real	may delete the interesting firms
Common in	regressions (1% / 99%)	obvious errors, other populations

ROA	N	mean	median
raw	30,280	−3.80	0.009
winsorized	30,280	−0.82	0.009
truncated	29,676	−0.49	0.009

Chunk 3.2

```
truncate <- function(x, p_low = 0.01, p_high = 0.99) {
  cuts <- quantile(x, c(p_low, p_high), type = 2, na.rm = TRUE)
  x[x < cuts[1] | x > cuts[2]] <- NA
  x }
```

Decide — and report

- Which variables, which cut-offs
- After the final sample is defined
- Pooled or by year?

▶️ **IN RSTUDIO** — **Part 3 · Chunks 3.2–3.4 + Tasks C1–C3: winsorize vs. truncate, boxplots, scatter plot**

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One concept, several proxies: how big is a firm?

Top 3 in 2014 by …	1	2	3
Market value	Apple	Exxon Mobil	Berkshire Hathaway
Total assets	Fannie Mae	JPMorgan Chase	Bank of America
Sales	Walmart	Exxon Mobil	Berkshire Hathaway
Employees	Walmart	Kelly Services	Yum! Brands

Correlation (2014)

- Pearson, raw values: **0.21 – 0.73**
- Spearman, ranks: **0.76 – 0.91**
- A few giant firms drive Pearson

Assets = capital invested · sales = activity · employees = organisation · market value = expected cash flows. **Choose the proxy that matches your mechanism.**

▶️ **IN RSTUDIO** — **Part 4 · Chunk 4.1 + Task D: compare size proxies — who are the ten largest firms?**

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Scaling and ranks: Lang & Lundholm's solution

Chunk 4.2 — the largest earnings surprises

```
# conm                     price_lag   surprise
# XXXX DORMANT SHELL CORP      0.001    835,995
# WESTPORT ENERGY HOLDINGS     0.0001   192,088
# median of all firm-years:             0.031
```

Chunk 4.3 — percentile rank within industry-year

```
cs |>
  group_by(sic2, fyear) |>
  mutate(surprise_r = (rank(surprise) - 1) /
                      (n() - 1))
```

Why ranks?

- Lowest firm in the industry-year = 0, highest = 1
- One extreme firm cannot dominate the results
- Industries become comparable — important for L&L, because each industry had its own FAF raters
- **But:** ranks lose the size of differences — and jump in small industries

▶️ **IN RSTUDIO** — **Part 4 · Chunks 4.2–4.3 + Task E: earnings surprise, winsorizing vs. ranks**

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Measuring disclosure today: completeness of ESRS reporting

Then: expert ratings (FAF)

- Analyst committees rate annual reports, other publications and investor relations
- Rich and judgment-based — but a perception, rated by the users themselves

Today: machine-readable reports (SRN)

- Completeness = share of 27 ESRS indicators found
- Median firm-year: **81%**; financials lowest (**63%**)
- Correlation with size (log revenues): only **0.13**

Is completeness a good proxy for informativeness?

- Materiality: not reporting land use can be correct · NOT_FOUND can be an extraction error · completeness says nothing about accuracy or clarity

▶️ **IN RSTUDIO** — **Part 4 · Chunks 4.4–4.5 + Task F: build the completeness proxy and question it**

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Design exercise: Lang & Lundholm (1996) today

Do firms with more informative sustainability disclosure attract more analysts?

Element	Your design
Concept X → proxy (data?)	
Concept Y → proxy (data?)	
Unit of observation, years	
Main confounder → control	
Weakest construct-validity link	
What would change your mind?	

Rules

- 15 minutes in pairs
- Text only — no code needed
- Name real data sources
- Present your **weakest link** in one sentence

▶️ **IN RSTUDIO** — **Part 5 · Fill in the design template — bonus: ask an AI tool for proxies and judge them**

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What you can do now — and what comes next

- ☐ I can draw Libby boxes and name the weakest link
- ☐ I can produce a descriptive table and read it like a referee
- ☐ I can build histograms, boxplots and scatter plots with ggplot2
- ☐ I know when to use logs, winsorizing or ranks — and report it
- ☐ I can compare several proxies for the same concept
- ☐ I can judge whether a proxy measures what I care about

Before 26 Nov

Commit and push your completed Tutorial 3

Thu 26 Nov

Tutorial 4 · Causal inference in practice

Fri 27 Nov

Tutorial 5 · Lab workshop

By 30 Nov

Lab paper approved

A perfectly estimated effect on the wrong proxy answers the wrong question.

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QUESTIONS?

See you on 26 November

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